

Muhammad Zulqarnain Abdullah

Taxila, Pakistan | muhammadzulqarnain1559@gmail.com | +92-3049322338

github.com/Mzaq1559 | linkedin.com/in/muhammad-zulqarnain-abdullah | mzaq1559.github.io/PortFolio | leetcode.com/u/Mzaq1559 | [Blog: mzaq1559.github.io/My-Learning-Diary](https://mzaq1559.github.io/My-Learning-Diary)

EDUCATION

University of Engineering and Technology (UET) Taxila

Bachelor of Science in Computer Science (In Progress) — 4th Semester **CGPA: 3.55/4.00**

TECHNICAL SKILLS

Languages: C++, Python, TypeScript, JavaScript, SQL, AVR Assembly

Full-Stack Development: React, Vite, Tailwind CSS, FastAPI, ASP.NET / .NET, REST APIs

Databases: PostgreSQL, Microsoft SQL Server, ChromaDB (vector database)

Computer Vision: OpenCV, YOLO (Ultralytics), ByteTrack / BoT-SORT multi-object tracking, object detection & tracking

AI/ML: Neural networks, backpropagation, neural networks from scratch, Transformers, PEFT/LoRA, RAG, LLM tool calling

Tools & DevOps: Git/GitHub, GitHub Actions, Docker, Azure App Service, Azure Container Registry, Linux (Ubuntu)

SELECTED PROJECTS

MediBook AI — AI Virtual Receptionist for Clinics | Alibaba Cloud AI Hackathon Pakistan 2026

GitHub: github.com/Mzaq1559/MEDIBOOK_AI

- Served as Project Lead for a 4-person team, owning system architecture, Docker Compose orchestration, database seeding, chat integration, and RAG integration for an AI-powered clinic receptionist with automated patient triage.
- Designed a full-stack architecture combining a React/TypeScript/Vite frontend, a FastAPI core backend and a separate FastAPI AI service, PostgreSQL for application data, and ChromaDB for retrieval over medical knowledge, with an LLM (via Groq) powering RAG-based triage conversations.

AutoVision — Real-Time Vehicle Intelligence & Traffic Analytics

GitHub: github.com/Mzaq1559/autovision-vehicle-intelligence

- Built a computer-vision pipeline chaining YOLOv8 (Ultralytics) object detection with ByteTrack/BoT-SORT multi-object tracking to detect, track, and analyze vehicles from video or webcam input.
- Implemented calibrated pixel-to-meter speed estimation, line-crossing vehicle counting, and directional/trajectory analysis, surfaced through a Streamlit dashboard with live Plotly analytics and a pytest suite covering the core logic.

AutoSolver — AI Delivery Dispatch System | Meituan 2026 International AI Hackathon

GitHub: github.com/Mzaq1559/Autosolver-dispatch

A real-time logistics platform for automated driver assignment, dynamic traffic simulation, and delivery tracking in a simulated Lahore delivery ecosystem.

- Built with a React/TypeScript/Vite frontend and a FastAPI/SQLite backend, synchronized live over WebSockets, with an asynchronous simulation engine driving driver movement and order lifecycle.
- Implemented an automated driver-assignment algorithm using Haversine distance and capacity-based scoring, a configurable time-warped simulation clock, and dynamic traffic effects, visualized on an interactive Leaflet map.

RCMS — Rental Car Management System | DBMS Coursework Project

GitHub: github.com/Mzaq1559/IDBS-Lab_Project

- Built a three-tier rental car system (FastAPI + pyodbc backend, vanilla JS frontend, Microsoft SQL Server) with fleet/reservation/invoice management, automated tax-inclusive invoicing, and double-booking prevention.

My Learning Diary — Self-Built Serverless Blogging Platform

GitHub: github.com/Mzaq1559/My-Learning-Diary

- Designed and built a serverless CMS (React, TypeScript, Vite, Tailwind CSS) that publishes and fetches Markdown-based posts directly from a GitHub repository via the Octokit REST API, with no backend server; identified and fixed a client-side secret-leak vulnerability in the CI build pipeline.

AI/ML FOCUS

- Areas of interest: Machine Learning & AI Research, Computer Vision, RAG & LLM Applications, Agentic AI Systems
- Implemented neural networks and a bigram language model from scratch while working through Andrej Karpathy's Micrograd and Makemore material
- Working through Hugging Face's structured NLP path: pipeline() API → AutoTokenizer/AutoModel → fine-tuning → PEFT/LoRA